

1. An automobile manufacturer is concerned about a fault in the braking mechanism of a particular model. The fault can, on rare occasions, cause a catastrophe at high speed. The distribution of the number of cars per year that will experience the catastrophe is a Poisson random variable with $\lambda = 5$.

(a) What is the probability that at most 3 cars per year will experience a catastrophe?
 (b) What is the probability that more than 1 car per year will experience a catastrophe?

2. Three cards are drawn without replacement from the 12 face cards (jacks, queens, and kings) of an ordinary deck of 52 playing cards. Let X be the number of kings selected and Y the number of jacks. Find

(a) the joint probability distribution of X and Y ;
 (b) Find the marginal probability of X dan Y
 (c) Check independency of X and Y

3. Consider the random variables X and Y with joint density function

$$f(x, y) = \begin{cases} x + y, & 0 \leq x \leq 1; 0 \leq y \leq 1 \\ 0, & \text{elsewhere.} \end{cases}$$

(a) Find the marginal distributions of X and Y .
 (b) Find $P(X > 0.25, Y > 0.5)$

4. X_1, X_2 , adalah variabel random saling bebas stokastik masing- masing dengan mean dan varians μ_1, σ_1^2 dan μ_2, σ_2^2 . Jika $Y = a_1X_1 + a_2X_2$ buktikan bahwa $\mu_Y = a_1\mu_1 + a_2\mu_2$ dan $\sigma_Y^2 = a_1^2\sigma_1^2 + a_2^2\sigma_2^2$.